

Retail Sales Forecasting

Project Overview

This project addresses inventory optimization for a retail company by building a machine learning system to forecast weekly product sales. Inconsistent sales predictions were leading to overstock and stockout situations, directly impacting revenue and customer satisfaction. The solution combines data preprocessing, missing-value imputation, exploratory analysis, and a time-series-aware forecasting model to deliver accurate, actionable sales predictions.

Problem Statement

A retail company wants to optimize inventory management by accurately forecasting sales for its products. Inconsistent sales predictions lead to overstock or stockouts, impacting overall revenue and customer satisfaction.

Objectives

- Develop a machine learning model to forecast product sales.
- Minimize stockouts and overstock situations by improving forecast accuracy.
- Provide insights into seasonality, trends, and external factors influencing sales.

Goals

- Achieve a forecasting accuracy of at least 90%.
- Reduce stockouts by 15% and overstock situations by 10%.
- Provide actionable insights for inventory optimization.

Project Workflow

1. Data Preprocessing

- Merged Store, Sales, and Features datasets on a common Store + Date key.
- Split Date into Day, Month, Year components.
- Converted negative Weekly_Sales values (data entry errors) to null for re-imputation.
- Created a binary 'markdown' flag indicating whether any promotional markdown was active.
- Identified null values across Markdown1–5, CPI, Unemployment, and Weekly_Sales.

2. Missing Value Imputation (Two Parallel Methods)

A sequential regression-based imputation pipeline (RandomForestRegressor, selected after comparing Decision Tree, Extra Trees, Random Forest, AdaBoost, Gradient Boosting, and XGBoost) was used to predict missing values in stages:

Method 1 - With Markdown features:

Markdown1–5 → CPI → Unemployment → Weekly_Sales

Method 2 - Without Markdown features:

CPI → Unemployment → Weekly_Sales (Markdown columns dropped due to high missingness/noise)

3. Outlier Analysis

- IQR-based detection flagged ~8.43% of rows as statistical outliers.
- Investigation showed these are not data errors but legitimate high-volume sales concentrated in large-format stores (Type 1, ~72% of outliers) and specific high-volume departments (38, 95, 92, 90, 40, 72).
- Outliers were retained; Store, Dept, Type, and Size features allow the model to learn scale differences naturally.

4. Time-Series Feature Engineering

To convert the static regression setup into a genuine time-series forecasting model, lag and rolling features were engineered per Store-Department combination:

- Sales_Lag1 - previous week's sales
- Sales_Lag2 - sales from two weeks prior
- Sales_RollMean4 - trailing 4-week rolling average (no data leakage; excludes current week)

A chronological (time-aware) train/test split was used (train: 2010–March 2013, test: April–July 2013) instead of a random split, to respect the temporal nature of the data.

5. Model Training & Evaluation

Final model: RandomForestRegressor trained on lag/rolling features plus store, department, and external economic/environmental features.

Results

Model Accuracy Comparison

Model	R ²	MAE	RMSE	Accuracy
Method 1 (with Markdown)	0.9737	1,434.25	3,650.16	97.37%
Method 2 (without Markdown)	0.9775	1,356.80	3,377.32	97.75%
Time-Series Model (final)	0.9927	723.88	1,774.80	94.82%

Note: Accuracy is reported using Weighted MAPE (WMAPE) rather than plain MAPE, since plain MAPE is distorted by rows with very small actual sales values (a known limitation of percentage-based error metrics in retail data).

Goal 1 ($\geq 90\%$ accuracy): Achieved - 94.82%

Stockout & Overstock Analysis

Forecast performance was benchmarked against a naive baseline ("last week's sales = this week's forecast," a common real-world heuristic).

	Stockout %	Overstock %
Naive Baseline	10.56%	11.51%
ML Model (final)	7.05%	21.17%
Reduction vs. Naive	33.30%	-83.87%

Goal 2 (Stockout $\geq 15\%$ reduction): Achieved - 33.30% (more than double the target)

Goal 2 (Overstock $\geq 10\%$ reduction): Not met - see Limitations below.

Feature Importance

Feature	Importance
Sales_Lag1	75.67%
Sales_RollMean4	19.16%
Sales_Lag2	1.33%
Dept	0.74%
Fuel_Price	0.59%
Temperature	0.52%
IsHoliday	0.45%
CPI	0.45%
Unemployment	0.44%
Store	0.30%
Size	0.29%
Type	0.05%

Recent sales history (last week + 4-week rolling average) accounts for ~95% of the model's predictive power - a hallmark of well-behaved retail time-series data.

Key Insights

- **Recency dominates:** Past sales (1-week lag + 4-week rolling average) are overwhelmingly the strongest predictors. Inventory systems should prioritize real-time tracking of recent sales over static store/economic attributes.
- **Strong seasonality:** Total sales spike 60–80% above baseline every November–December (holiday shopping season), followed by a sharp correction in January - a consistent pattern across all three years in the dataset.
- **Outliers are demand signals, not noise:** High-value transactions cluster in large-format (Type 1) stores and a small set of high-volume departments - these should be treated as expected peak demand, not anomalies to be cleaned away.
- **Stockout risk substantially reduced:** The model more than doubles the target reduction in stockout events compared to naive forecasting, directly addressing lost-sales risk.
- **Stockout–overstock trade-off is fundamental, not fixable by simple calibration:** Prediction bias varies systematically by store type and department (confirmed by both additive and proportional calibration experiments - no single global correction satisfied both the stockout and overstock targets simultaneously).

Limitations & Recommendations

- Overstock target not met with a single global model. The forecasting bias is heterogeneous across store types and departments - a uniform adjustment that reduces overstock always increases stockout risk, and vice versa (the classic "newsvendor" trade-off in inventory theory).
- **Recommendation:** Implement segment-specific safety stock buffers rather than one global calibration - e.g., lower safety buffers for Type 1 (large) stores, where the model tends to over-predict, and higher buffers for smaller stores/departments prone to under-prediction.
- **Business prioritization:** Given that stockouts typically carry a higher cost (lost sales, customer dissatisfaction) than overstock (holding cost), the final model was tuned to prioritize stockout reduction.
- **Future work:** A quantile regression or per-segment calibration approach (tested at a preliminary level in this project) could be extended with more granular segment definitions to close the overstock gap without sacrificing stockout performance.

Tech Stack

- Language: Python
- Data Handling: pandas, NumPy
- Visualization: matplotlib, seaborn
- Modeling: scikit-learn (RandomForestRegressor, DecisionTreeRegressor, ExtraTreesRegressor, AdaBoostRegressor, GradientBoostingRegressor), XGBoost
- Persistence: pickle

Deliverables

- model/final_timeseries_model.pkl - final trained forecasting model
- model/model1_markdown.pkl, model/model2.pkl - intermediate Weekly_Sales models (Method 1 & 2)
- outputs/model_comparison.csv - accuracy metrics across all models
- outputs/stockout_overstock_summary.csv - inventory impact analysis

- outputs/feature_importance.csv - feature importance rankings
- outputs/key_insights.txt - summarized insights
- checkpoint/full_checkpoint_v2.pkl - full pipeline state (for reproducibility without rerunning)

Project Scope Completion

Scope Item	Status
Data analysis and preprocessing	Complete
Feature engineering and model development	Complete
Model evaluation and optimization	Complete
Insights generation and recommendations	Complete